Project wireless

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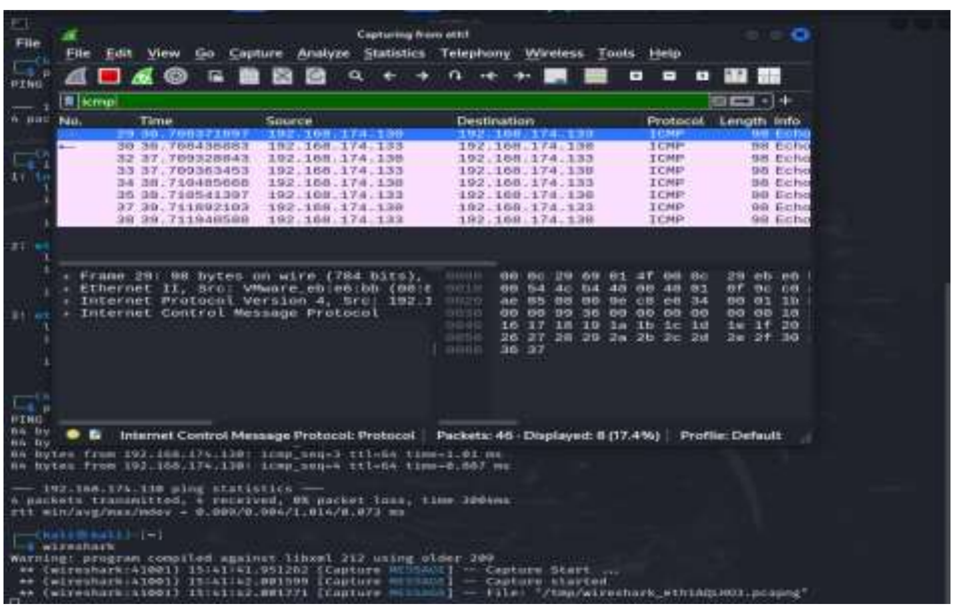
Ping ubuntu from kali

```
┌──(kali㉿kali)-[~]
└─$ ping -c 4 192.168.174.130
PING 192.168.174.130 (192.168.174.130) 56(84) bytes of data.
64 bytes from 192.168.174.130: icmp_seq=1 ttl=64 time=0.908 ms
64 bytes from 192.168.174.130: icmp_seq=2 ttl=64 time=0.809 ms
64 bytes from 192.168.174.130: icmp_seq=3 ttl=64 time=1.01 ms
64 bytes from 192.168.174.130: icmp_seq=4 ttl=64 time=0.887 ms

—— 192.168.174.130 ping statistics ——
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.809/0.904/1.014/0.073 ms
```

Ping kali from ubuntu

```
cyberarena@ubuntu:~$ ping -c 4 192.168.174.133
PING 192.168.174.133 (192.168.174.133) 56(84) bytes of data.
64 bytes from 192.168.174.133: icmp_seq=1 ttl=64 time=0.815 ms
64 bytes from 192.168.174.133: icmp_seq=2 ttl=64 time=0.757 ms
64 bytes from 192.168.174.133: icmp_seq=3 ttl=64 time=0.687 ms
64 bytes from 192.168.174.133: icmp_seq=4 ttl=64 time=0.621 ms

--- 192.168.174.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.621/0.720/0.815/0.072 ms
cyberarena@ubuntu:~$
```

Icmp in wireshark



Week 2

```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.174.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:44 EDT
Nmap scan report for 192.168.174.1
Host is up (0.00065s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 192.168.174.130
Host is up (0.00032s latency).
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Nmap scan report for 192.168.174.254
Host is up (0.00026s latency).
MAC Address: 00:50:56:F7:6B:7A (VMware)
Nmap scan report for 192.168.174.133
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 15.14 seconds
```

Commented [s1]: Sn for host discovery

192.168.174.130 → Ubuntu target

192.168.174.133 → Kali attacker

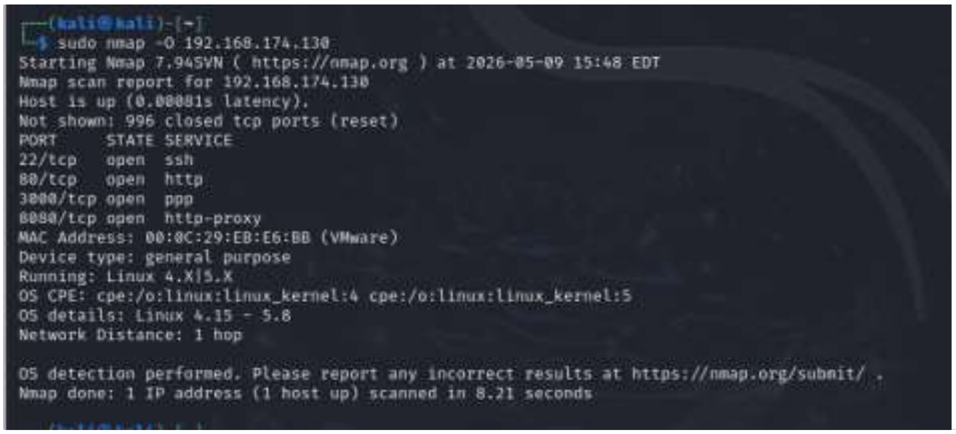
```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.174.130
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:46 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00054s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
MAC Address: 00:0C:29:EB:E6:BB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.05 seconds
```

Commented [s2]: -Ss stealth scan

This performs a SYN scan against the Ubuntu target for ports 1–1000.

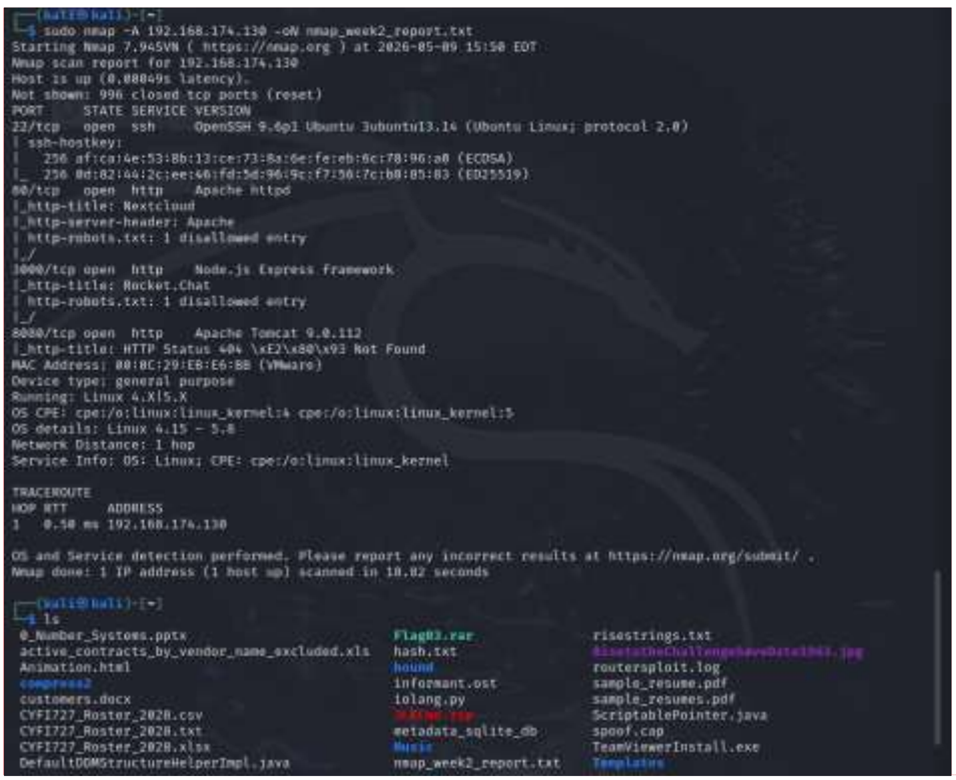
```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.174.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:47 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00036s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
80/tcp   open  http    Apache httpd
3000/tcp open  http    Node.js Express framework
8080/tcp open  http    Apache Tomcat 9.0.112
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.64 seconds
```

Commented [s3]: -Sv for service version



```
$ sudo nmap -O 192.168.174.130
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:48 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00081s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
80/tcp   open  http
3000/tcp open  ppp
8080/tcp open  http-proxy
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.21 seconds
```

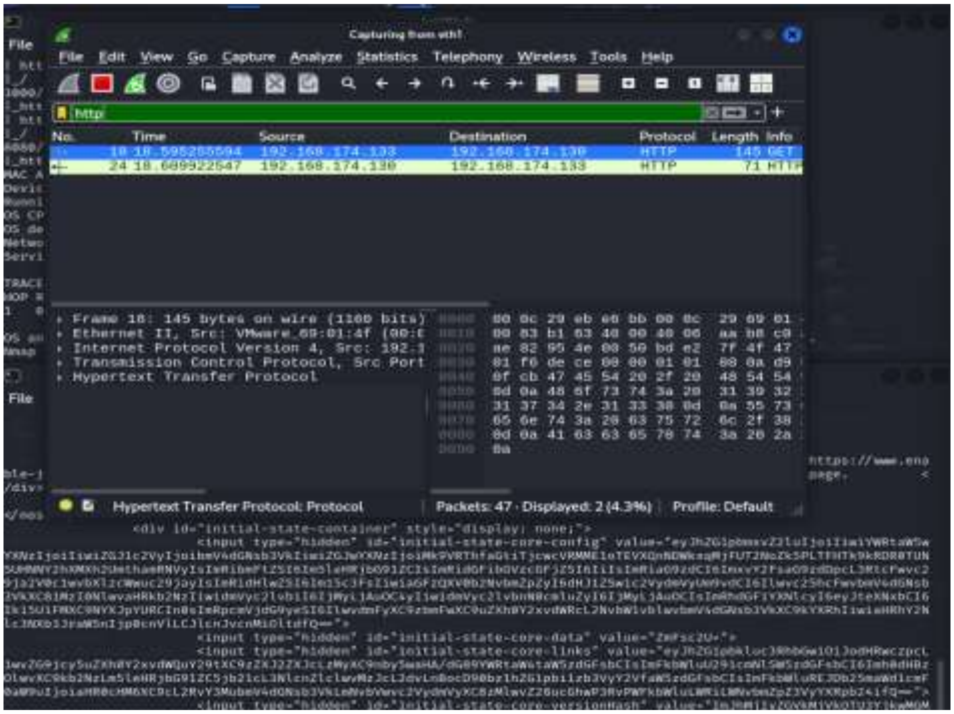
Commented [s4]: -O for operating system

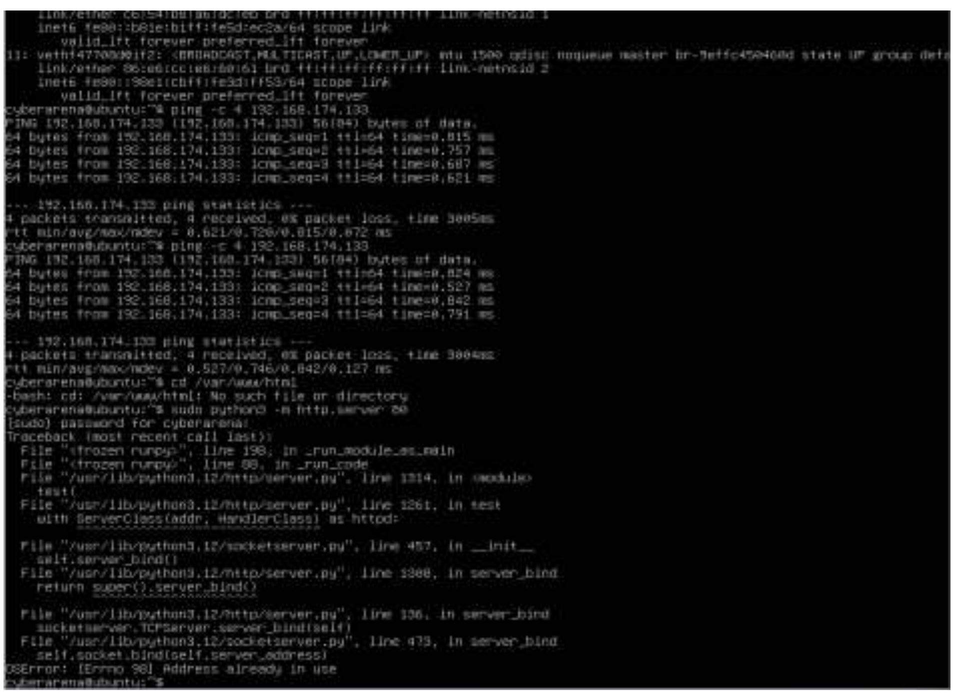


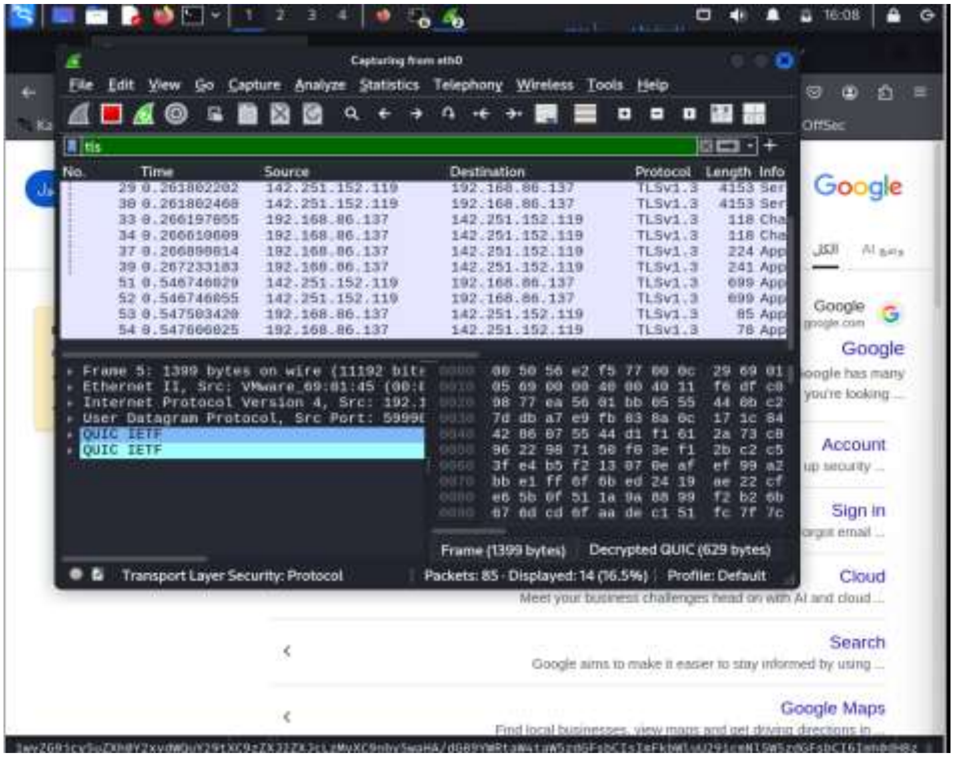
```
$ sudo nmap -A 192.168.174.130 -oN nmap_week2_report.txt
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 15:50 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00049s latency).
Not shown: 996 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
22/tcp   open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 af:ca:4e:53:8b:13:ce:73:8a:6e:fe:eb:6c:78:96:a0 (ECDSA)
|_  256 0d:82:44:2c:ee:46:fd:5d:96:9c:f7:56:7c:b0:05:83 (ED25519)
80/tcp   open  http    Apache httpd
|_http-title: Nextcloud
|_http-server-header: Apache
| http-robots.txt: 1 disallowed entry
|_/
3000/tcp open  http    Node.js Express framework
|_http-title: Rocket.Chat
| http-robots.txt: 1 disallowed entry
|_/
8080/tcp open  http    Apache Tomcat 9.0.112
|_http-title: HTTP Status 404 \xE2\x80\x93 Not Found
MAC Address: 00:0C:29:EB:E6:BB (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT     ADDRESS
1   0.50 ms 192.168.174.130

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.02 seconds

$ ls
0_Number_Systems.pptx                        Flag03.rar              risestrings.txt
active_contracts_by_vendor_name_excluded.xls hash.txt
Animation.html                               hound                   routersploit.log
compress2                                    informant.ost           sample_resume.pdf
customers.docx                               iolang.py               sample_resumes.pdf
CYFI727_Roster_2020.csv                                              ScriptablePointer.java
CYFI727_Roster_2020.txt                      metadata_sqlite_db      spoof.cap
CYFI727_Roster_2020.xlsx                     Music                   TeamViewerInstall.exe
DefaultDDMStructureHelperImpl.java           nmap_week2_report.txt   Templates
```

Commented [s5]: -A AGGRESSIVE -oN save out nmap in text file

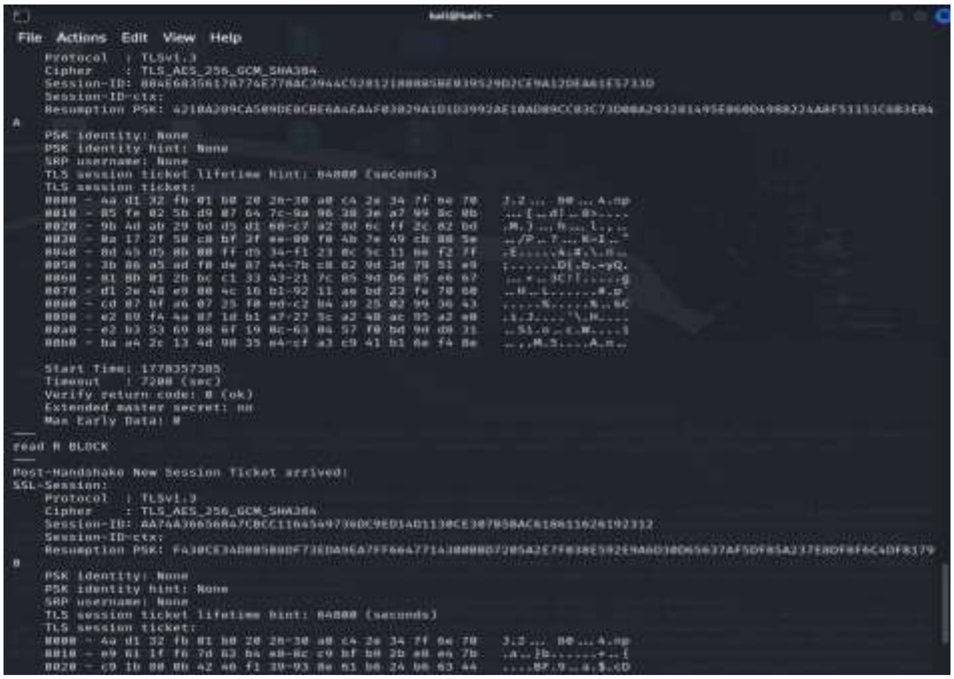
Week 3


Capturing from eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
Frame 18: 145 bytes on wire (1160 bits)
Ethernet II, Src: VMware_69:01:4f
Internet Protocol Version 4, Src: 192.1
Transmission Control Protocol, Src Port
Hypertext Transfer Protocol
Hypertext Transfer Protocol: Protocol
Packets: 47 · Displayed: 2 (4.3%)
Profile: Default

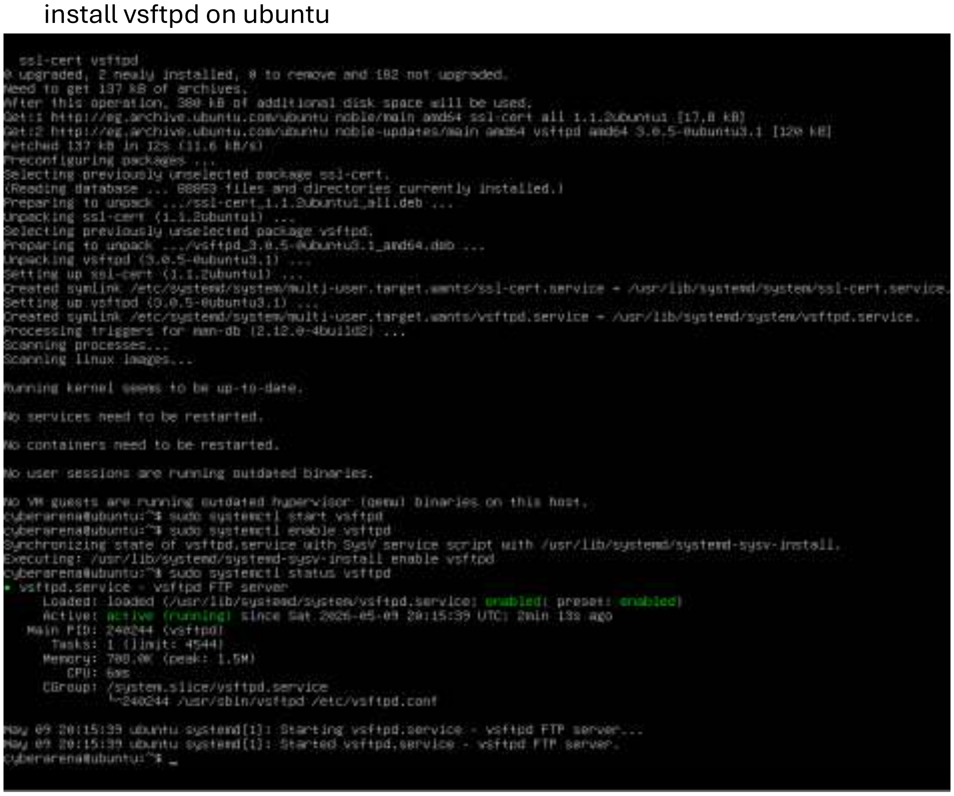

cyberarena@ubuntu:~$ ping -c 4 192.168.174.133
--- 192.168.174.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss
cyberarena@ubuntu:~$ sudo python3 -m http.server 80
Traceback (most recent call last):
OSError: [Errno 98] Address already in use

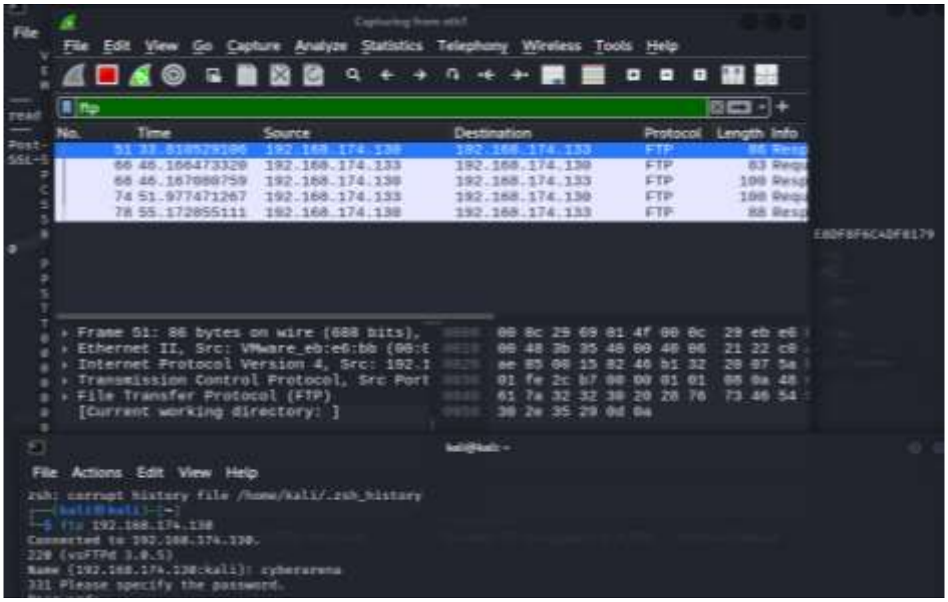


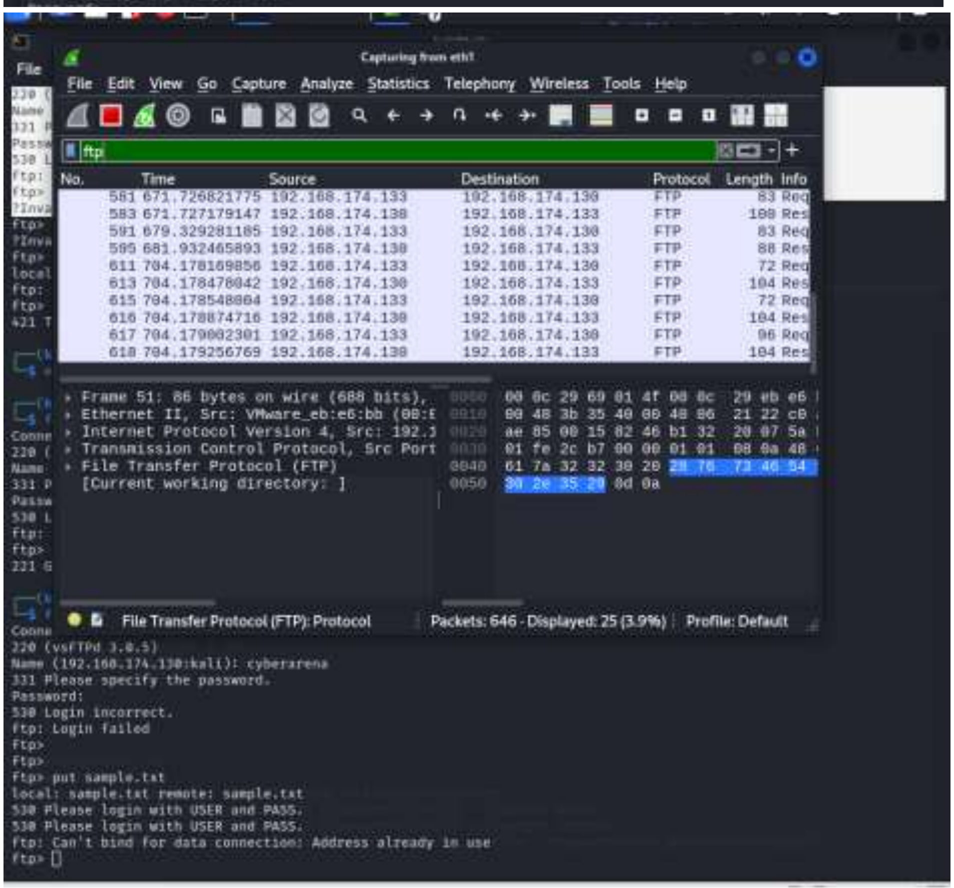
Week 4

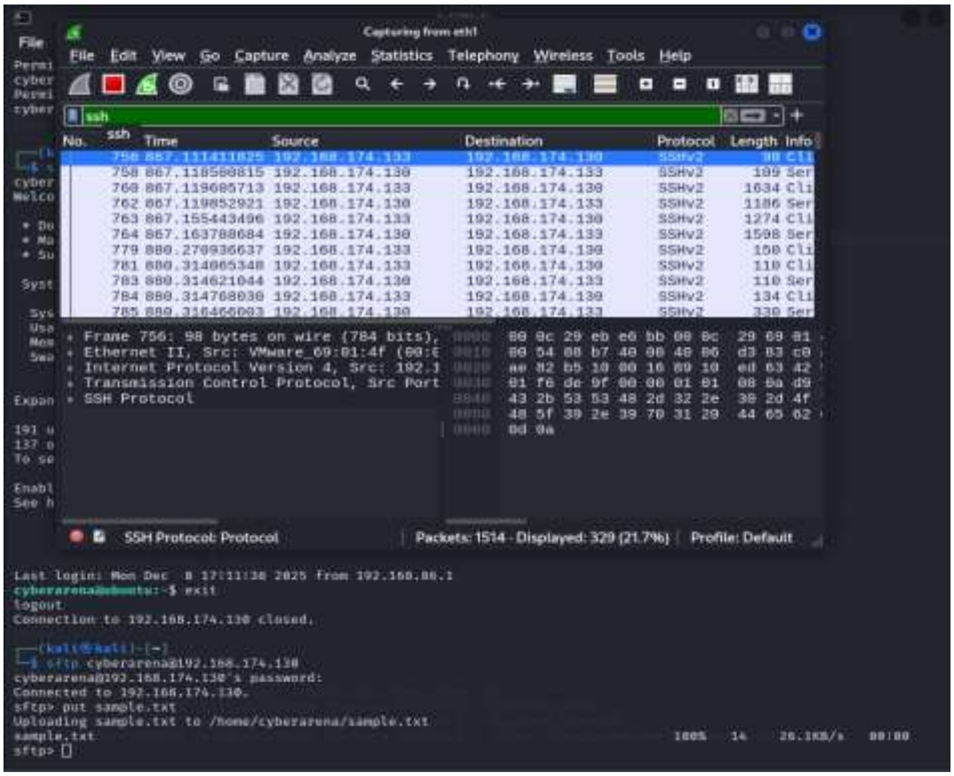


install vsftpd on ubuntu



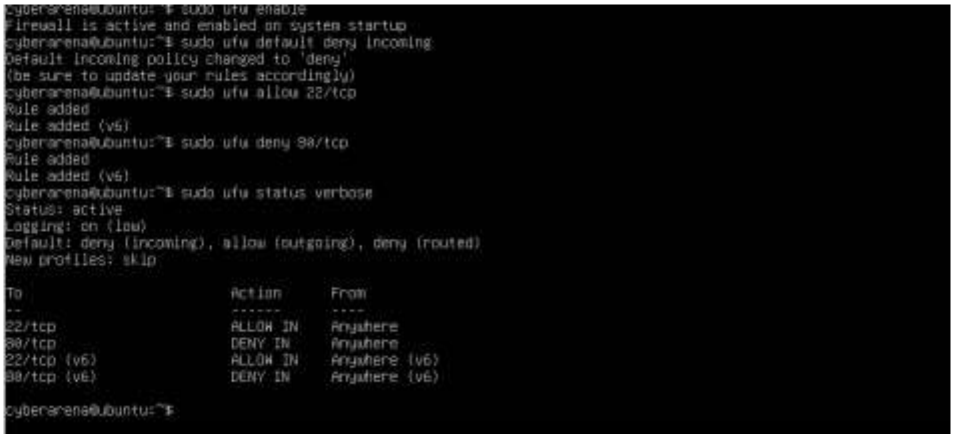

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ftp
No. Time Source Destination Protocol Length Info
Frame 51: 86 bytes on wire (688 bits),
Ethernet II, Src: VMware_eb:e6:bb
Internet Protocol Version 4, Src: 192.1
Transmission Control Protocol, Src Port
File Transfer Protocol (FTP)
[Current working directory:]
File Actions Edit View Help

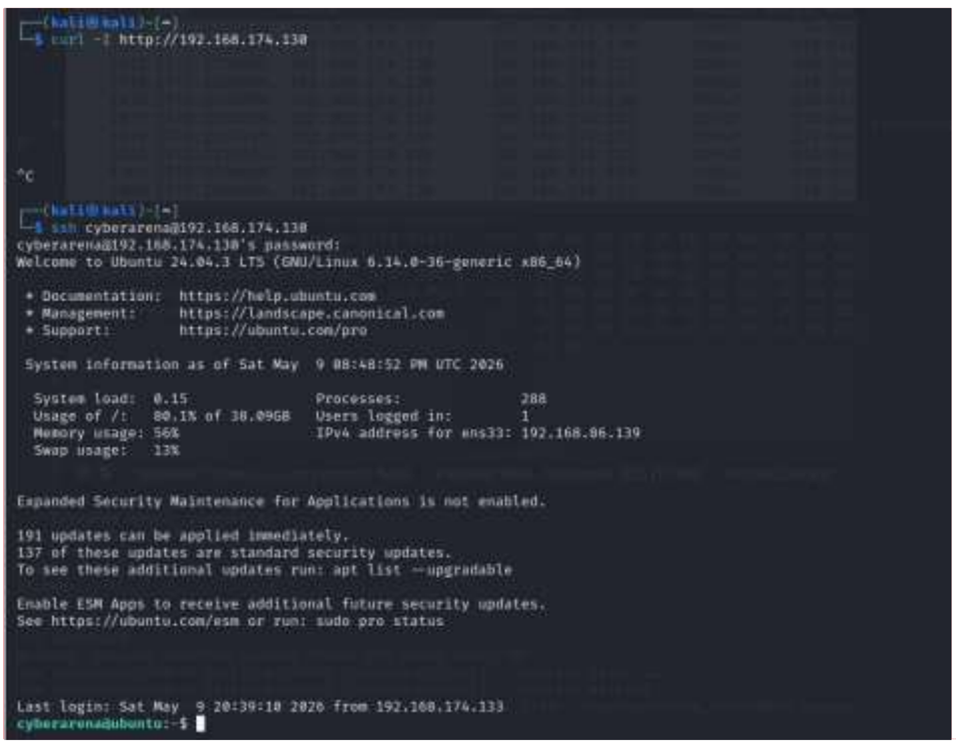

Capturing from eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
ftp
No. Time Source Destination Protocol Length Info
581 671.726821775 192.168.174.133 192.168.174.130 FTP 83 Req
583 671.727179147 192.168.174.130 192.168.174.133 FTP 100 Res
591 679.329281185 192.168.174.133 192.168.174.130 FTP 83 Req
595 681.932465893 192.168.174.130 192.168.174.133 FTP 88 Res
611 704.178169856 192.168.174.133 192.168.174.130 FTP 72 Req
613 704.178478042 192.168.174.130 192.168.174.133 FTP 104 Res
615 704.178548004 192.168.174.133 192.168.174.130 FTP 72 Req
616 704.178874716 192.168.174.130 192.168.174.133 FTP 104 Res
617 704.179002301 192.168.174.133 192.168.174.130 FTP 96 Req
618 704.179256769 192.168.174.130 192.168.174.133 FTP 104 Res
Frame 51: 86 bytes on wire (688 bits),
Ethernet II, Src: VMware_eb:e6:bb
Internet Protocol Version 4, Src: 192.1
Transmission Control Protocol, Src Port
File Transfer Protocol (FTP)
[Current working directory:]
File Transfer Protocol (FTP): Protocol
Packets: 646 · Displayed: 25 (3.9%)
Profile: Default
220 (vsFTPd 3.0.5)
Name (192.168.174.130:kali): cyberarena
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp>
ftp>
ftp> put sample.txt
local: sample.txt remote: sample.txt
530 Please login with USER and PASS.
530 Please login with USER and PASS.
ftp: Can't bind for data connection: Address already in use
ftp>



FTP	SFTP
Plaintext	Encrypted
Credentials visible	Credentials hidden
Readable commands	Encrypted SSH packets

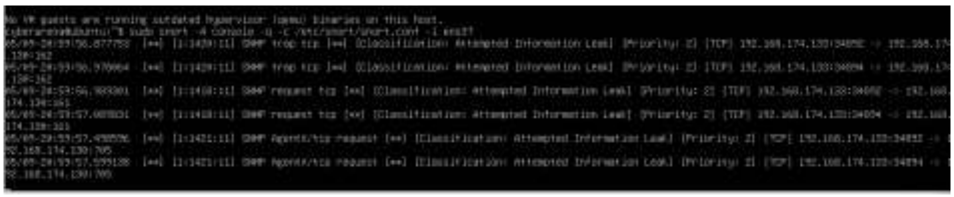
Week 5

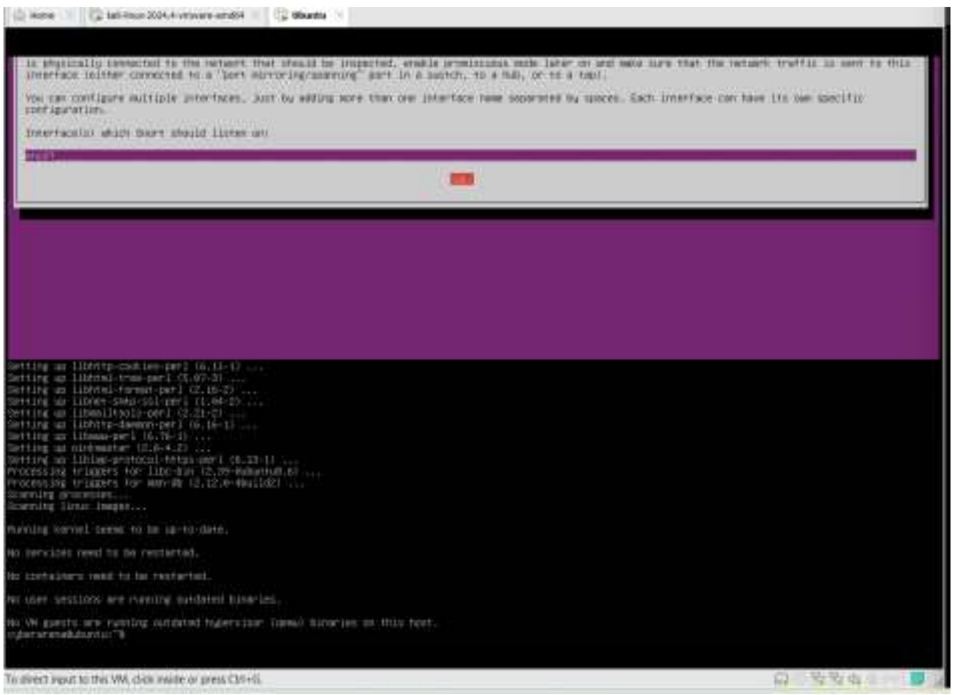




Commented [s6]: HTTP (curl) was blocked by UFW SSH login still succeeded

Week 6



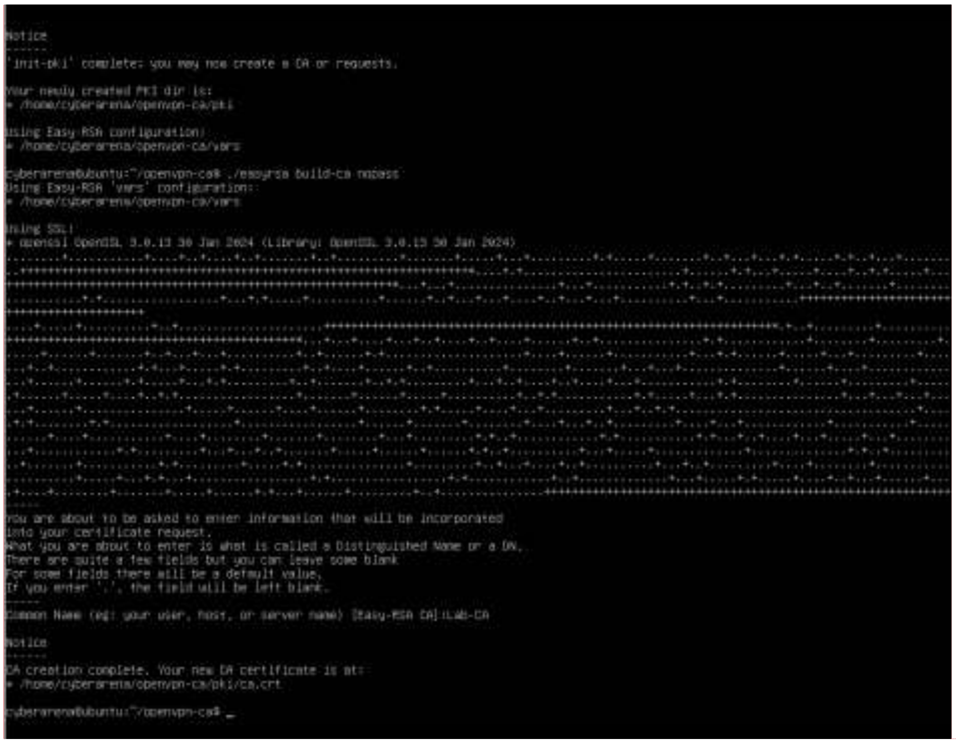


```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.174.130
[sudo] password for kali:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-09 16:59 EDT
Nmap scan report for 192.168.174.130
Host is up (0.00031s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
22/tcp open  ssh
MAC Address: 00:0C:29:EB:E6:BB (VMware)

Nmap done: 1 IP address (1 host up) scanned in 11.48 seconds

┌──(kali㉿kali)-[~]
└─$
```

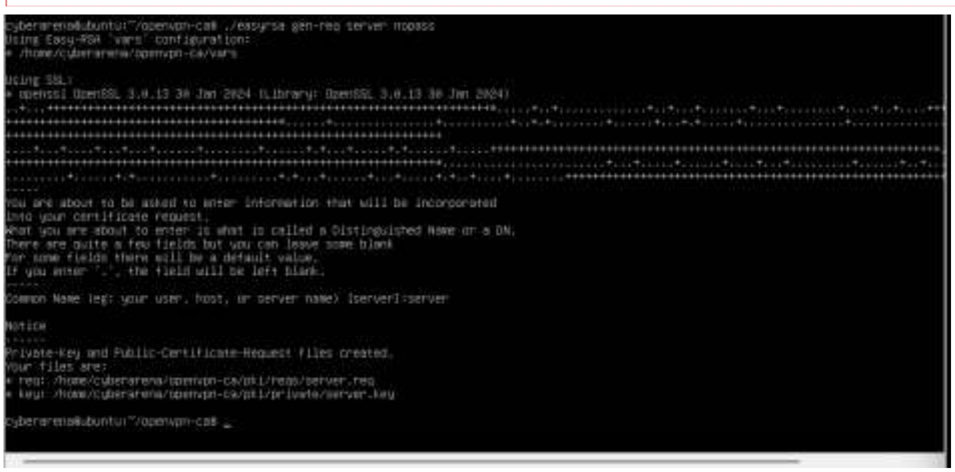
Week 8

Commented [s7]: OpenVPN and Easy-RSA installed successfully.

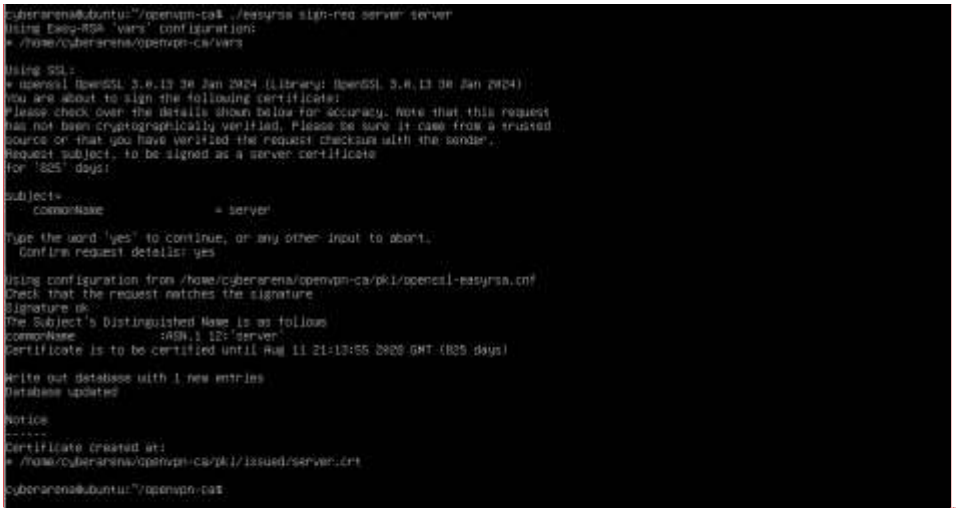
```
Preparing to unpack .../1-libpcsclite1_2.0.3-1build1_amd64.deb ...
Unpacking libpcsclite1:amd64 (2.0.3-1build1) ...
Selecting previously unselected package pcscd.
Preparing to unpack .../2-pcscd_2.0.3-1build1_amd64.deb ...
Unpacking pcscd (2.0.3-1build1) ...
Selecting previously unselected package libeac3:amd64.
Preparing to unpack .../3-libeac3_1.1.2+ds+git20220117+453c3d6b03a0-1.1build2_amd64.deb ...
Unpacking libeac3:amd64 (1.1.2+ds+git20220117+453c3d6b03a0-1.1build2) ...
Selecting previously unselected package libpkcs11-helper1t64:amd64.
Preparing to unpack .../4-libpkcs11-helper1t64_1.29.0-2.1build2_amd64.deb ...
Unpacking libpkcs11-helper1t64:amd64 (1.29.0-2.1build2) ...
Selecting previously unselected package opensc-pkcs11:amd64.
Preparing to unpack .../5-opensc-pkcs11_0.25.0~rc1-1ubuntu0.2_amd64.deb ...
Unpacking opensc-pkcs11:amd64 (0.25.0~rc1-1ubuntu0.2) ...
Selecting previously unselected package opensc.
Preparing to unpack .../6-opensc_0.25.0~rc1-1ubuntu0.2_amd64.deb ...
Unpacking opensc (0.25.0~rc1-1ubuntu0.2) ...
Selecting previously unselected package openvpn.
Preparing to unpack .../7-openvpn_2.6.19-0ubuntu0.24.04.1_amd64.deb ...
Unpacking openvpn (2.6.19-0ubuntu0.24.04.1) ...
Selecting previously unselected package easy-rsa.
Preparing to unpack .../8-easy-rsa_3.1.7-2_all.deb ...
Unpacking easy-rsa (3.1.7-2) ...
Setting up libccid (1.5.5-1) ...
Setting up libeac3:amd64 (1.1.2+ds+git20220117+453c3d6b03a0-1.1build2) ...
Setting up opensc-pkcs11:amd64 (0.25.0~rc1-1ubuntu0.2) ...
Setting up libpcsclite1:amd64 (2.0.3-1build1) ...
Setting up libpkcs11-helper1t64:amd64 (1.29.0-2.1build2) ...
Setting up easy-rsa (3.1.7-2) ...
Setting up openvpn (2.6.19-0ubuntu0.24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/openvpn.service → /usr/lib/systemd/system/openvpn.service.
Setting up opensc (0.25.0~rc1-1ubuntu0.2) ...
Setting up pcscd (2.0.3-1build1) ...
Created symlink /etc/systemd/system/sockets.target.wants/pcscd.socket → /usr/lib/systemd/system/pcscd.socket.
pcscd.service is a disabled or a static unit, not starting it.
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
cyberarena@ubuntu:~$
```


```
Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /home/cyberarena/openvpn-ca/pki

Using Easy-RSA configuration:
* /home/cyberarena/openvpn-ca/vars

cyberarena@ubuntu:~/openvpn-ca$ ./easyrsa build-ca nopass
Using Easy-RSA 'vars' configuration:
* /home/cyberarena/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:Lab-CA

Notice
------
CA creation complete. Your new CA certificate is at:
* /home/cyberarena/openvpn-ca/pki/ca.crt

cyberarena@ubuntu:~/openvpn-ca$ _
```

Commented [s8]: PKI initialized
CA certificate generated
OpenVPN trust infrastructure ready

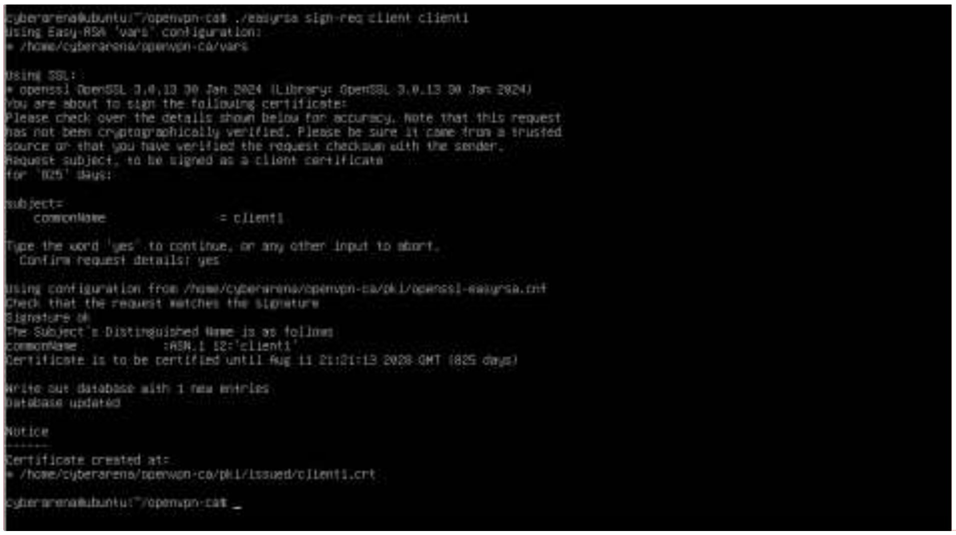

```
cyberarena@ubuntu:~/openvpn-ca$ ./easyrsa gen-req server nopass
Using Easy-RSA 'vars' configuration:
* /home/cyberarena/openvpn-ca/vars

Using SSL:
* openssl OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [server]:server

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /home/cyberarena/openvpn-ca/pki/reqs/server.req
* key: /home/cyberarena/openvpn-ca/pki/private/server.key

cyberarena@ubuntu:~/openvpn-ca$ _
```

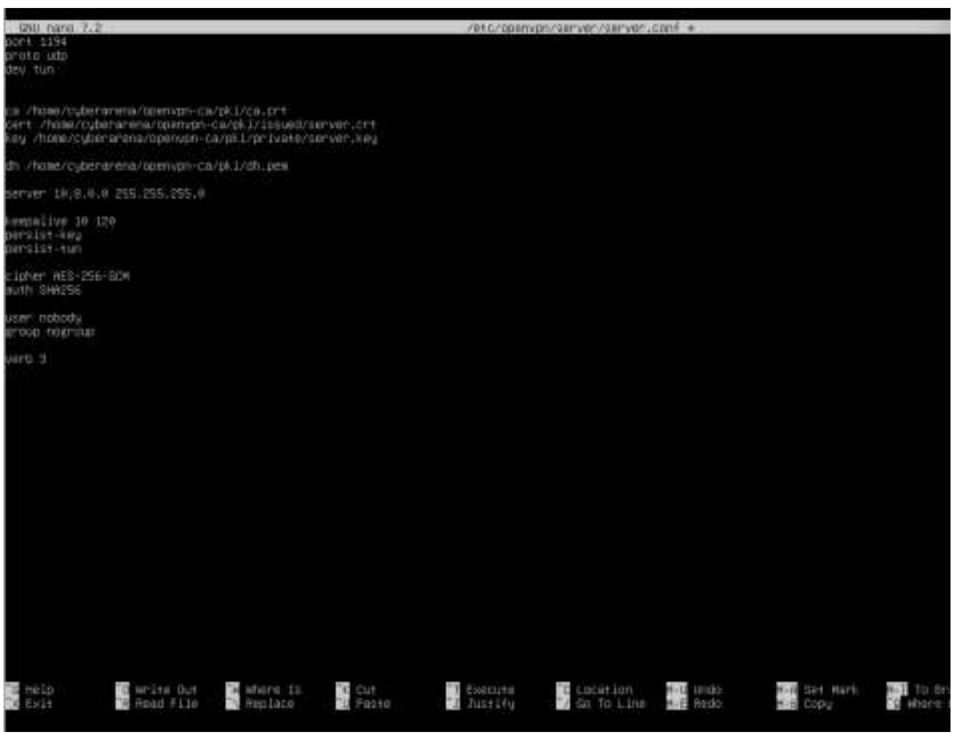
Commented [s9]: Generate Server Certificate



Commented [s10]: Sign Server Certificate

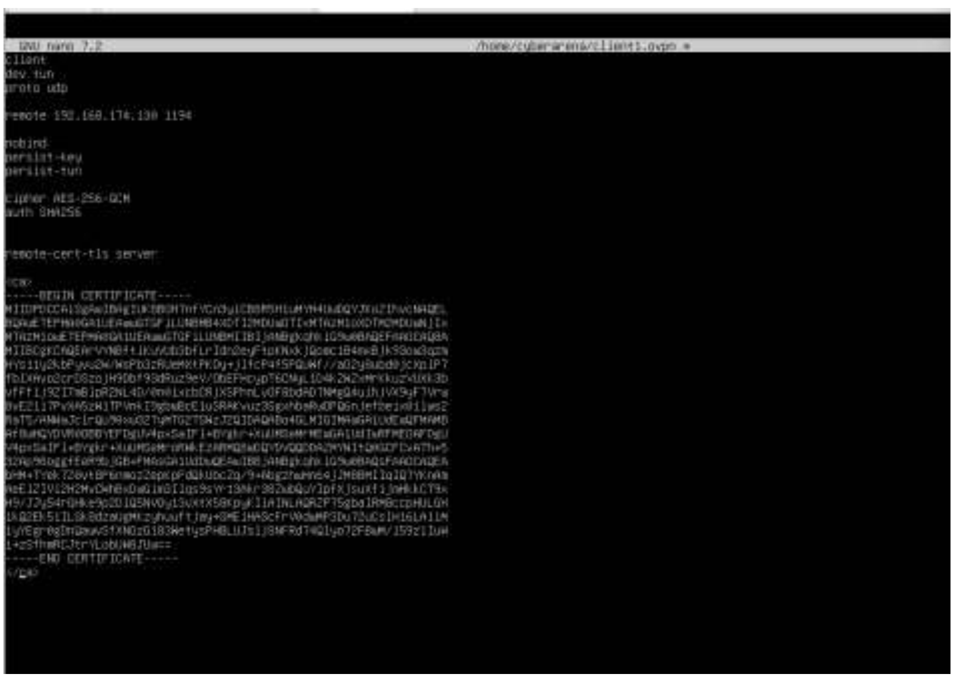
Commented [s11]: Generate Client Certificate

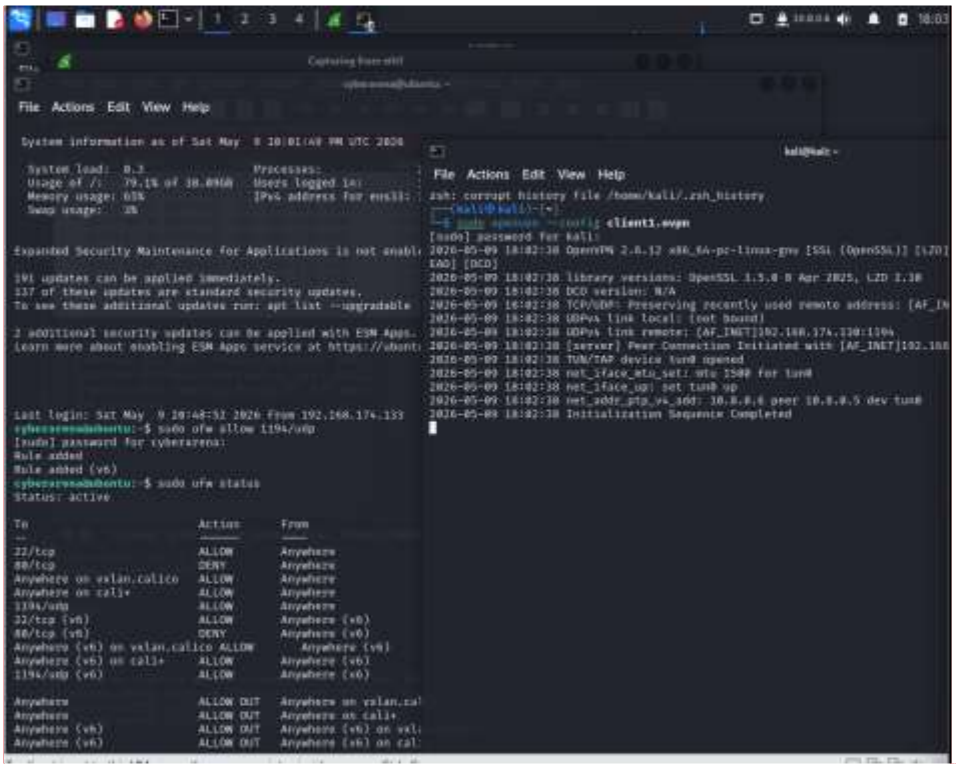


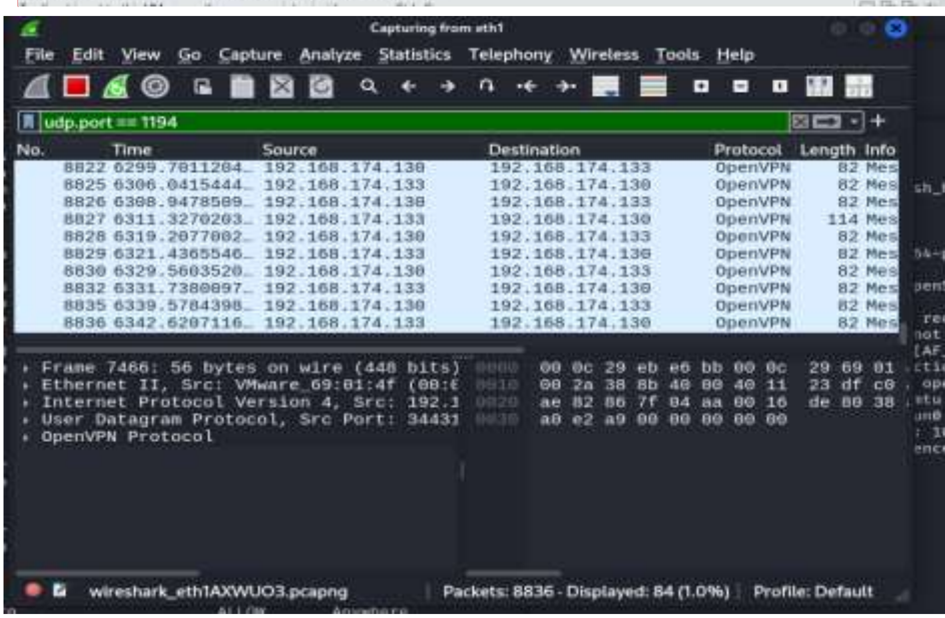


Commented [s12]: Sign Client Certificate

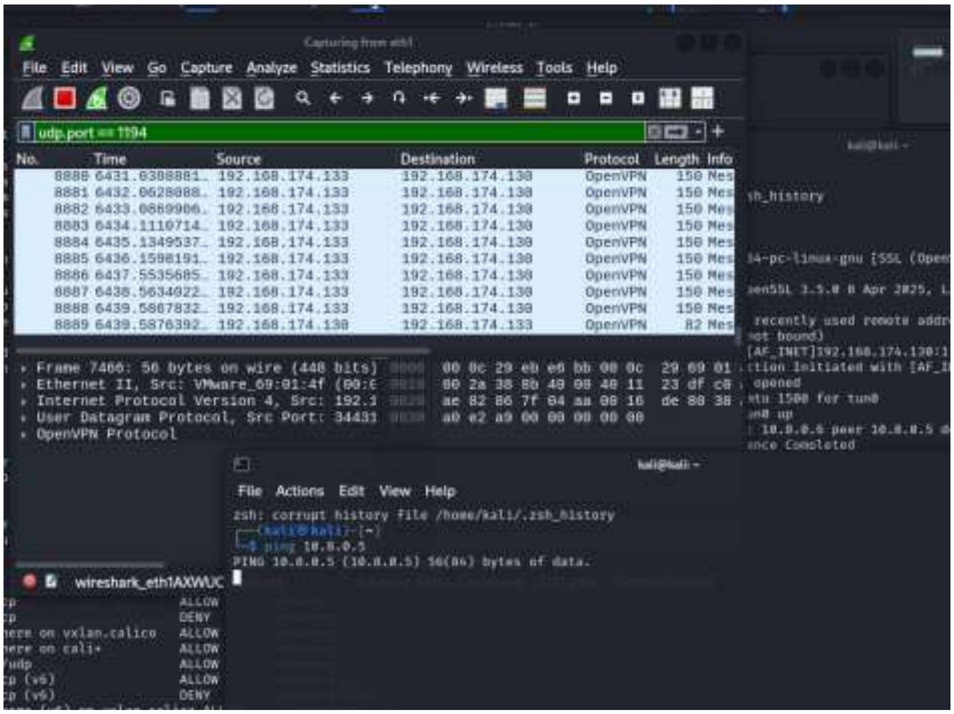

cyberarena@ubuntu:~/openvpn-ca$ sudo openvpn --config /etc/openvpn/server/server.conf
2026-05-09 21:37:23 WARNING: --topology net30 support for server configs with IPv4 pools will be removed in a future release. Please migrate it as soon as possible.
2026-05-09 21:37:23 Note: NOT using '--topology subnet' disables data channel offload.
2026-05-09 21:37:23 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-05-09 21:37:23 DCO version: N/A
2026-05-09 21:37:23 Diffie-Hellman initialized with 2048 bit key
2026-05-09 21:37:23 TUN/TAP device tun0 opened
2026-05-09 21:37:23 net_iface_mtu_set: mtu 1500 for tun0
2026-05-09 21:37:23 net_iface_up: set tun0 up
2026-05-09 21:37:23 Could not determine IPv4/IPv6 protocol. Using AF_INET
2026-05-09 21:37:23 UDPv4 link remote: [AF_UNSPEC]
2026-05-09 21:37:23 UID set to nobody
2026-05-09 21:37:23 GID set to nogroup
2026-05-09 21:37:23 Capabilities retained: CAP_NET_ADMIN
2026-05-09 21:37:23 MULTI: multi_init called, r=256 v=256
2026-05-09 21:37:23 IFCONFIG POOL IPv4: base=10.8.0.4 size=62
2026-05-09 21:37:23 Initialization Sequence Completed


GNU nano 7.2
/home/cyberarena/client1.ovpn *
client
dev tun
proto udp
remote 192.168.174.130 1194
nobind
persist-key
persist-tun
cipher AES-256-GCM
auth SHA256
remote-cert-tls server
<ca>
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
</ca>

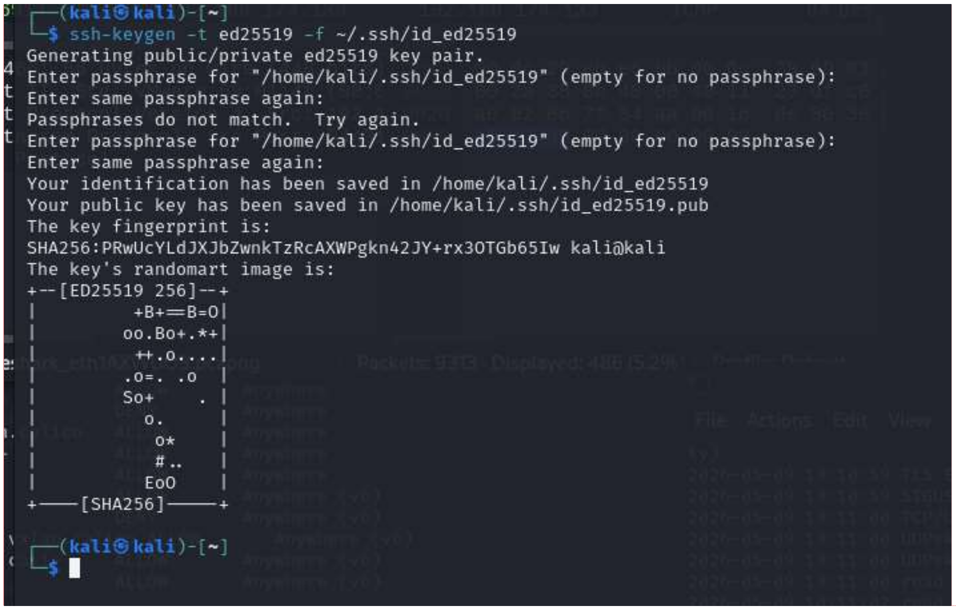




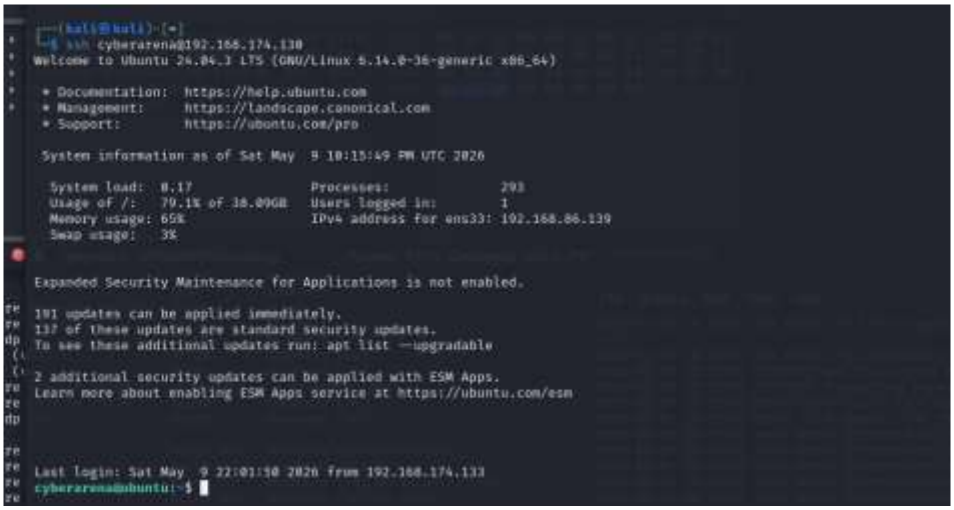
Commented [s13]: Kali connected to the OpenVPN server,
encrypted tunnel established,
tun0 interface created,
VPN IPs assigned successfully

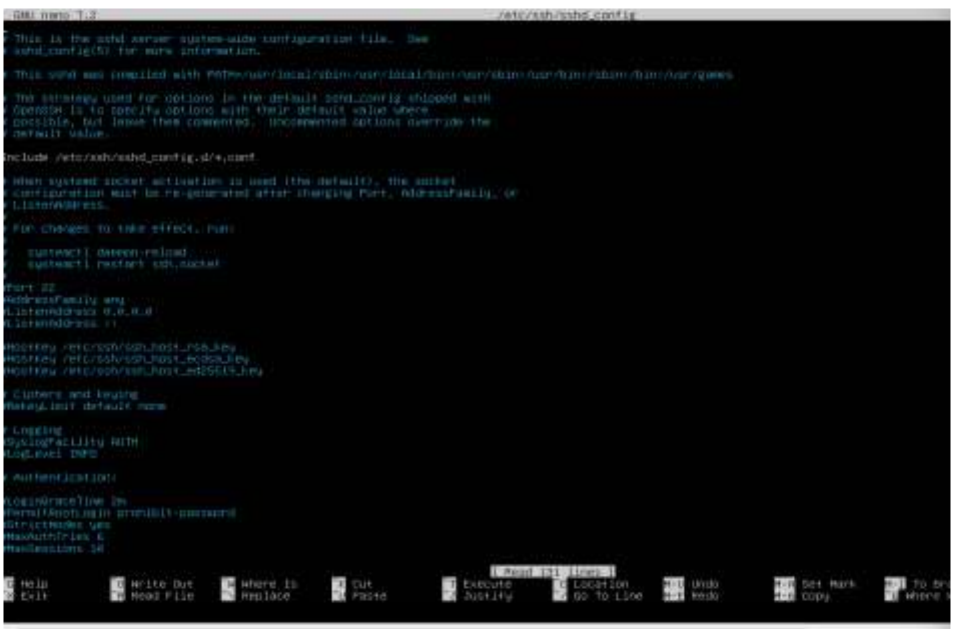

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp.port == 1194
No. Time Source Destination Protocol Length Info
8880 6431.0388881.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8881 6432.0628088.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8882 6433.0869906.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8883 6434.1110714.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8884 6435.1349537.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8885 6436.1598191.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8886 6437.5535685.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8887 6438.5634022.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8888 6439.5867832.. 192.168.174.133 192.168.174.130 OpenVPN 150 Mes
8889 6439.5876392.. 192.168.174.130 192.168.174.133 OpenVPN 82 Mes
Frame 7466: 56 bytes on wire (448 bits)
Ethernet II, Src: VMware_69:01:4f
Internet Protocol Version 4, Src: 192.1
User Datagram Protocol, Src Port: 34431
OpenVPN Protocol
wireshark_eth1AXWUC
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
ping 10.8.0.5
PING 10.8.0.5 (10.8.0.5) 56(84) bytes of data.
sh_history
recently used remote addr
not bound)
ction Initiated with [AF_I
opened
mtu 1500 for tun0
un0 up
10.8.0.6 peer 10.8.0.5
nce Completed
ALLOW
DENY
here on vxlan.calico ALLOW
here on cali+ ALLOW
/udp ALLOW
p (v6) ALLOW
p (v6) DENY

Week 9

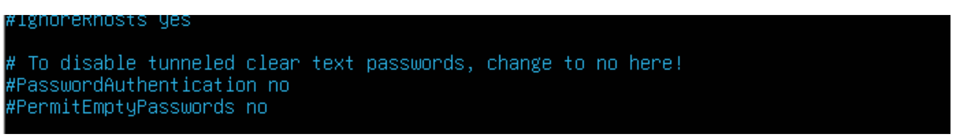


Commented [s14]: Generate SSH Key on Kali now is logged without password





Chanage to




cyberarena@ubuntu:~$ sudo systemctl reload apache2
cyberarena@ubuntu:~$


PORT STATE SERVICE VERSION
ssh-hostkey:
Device type: general purpose|storage-misc
TRACEROUTE
HOP RTT ADDRESS